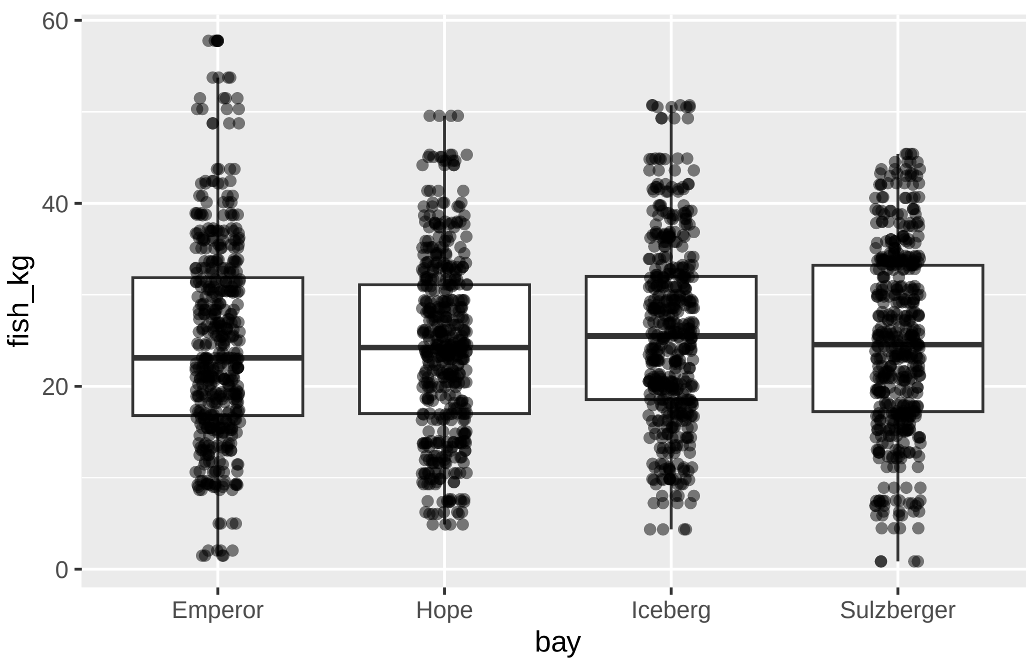
Module 3, Assignment 2: Expected Outputs

3.

```
# A tibble: 4 x 3
  bay         mean_fish sd_fish
  <chr>           <dbl>   <dbl>
1 Emperor          24.9   10.9
2 Hope             24.3    9.73
3 Iceberg          25.8    9.94
4 Sulzberger       24.9    9.98
```

4.



6.

```
              Df Sum Sq Mean Sq F value Pr(>F)
bay            3    542   180.6   1.751  0.155
Residuals   1916 197625   103.1
```

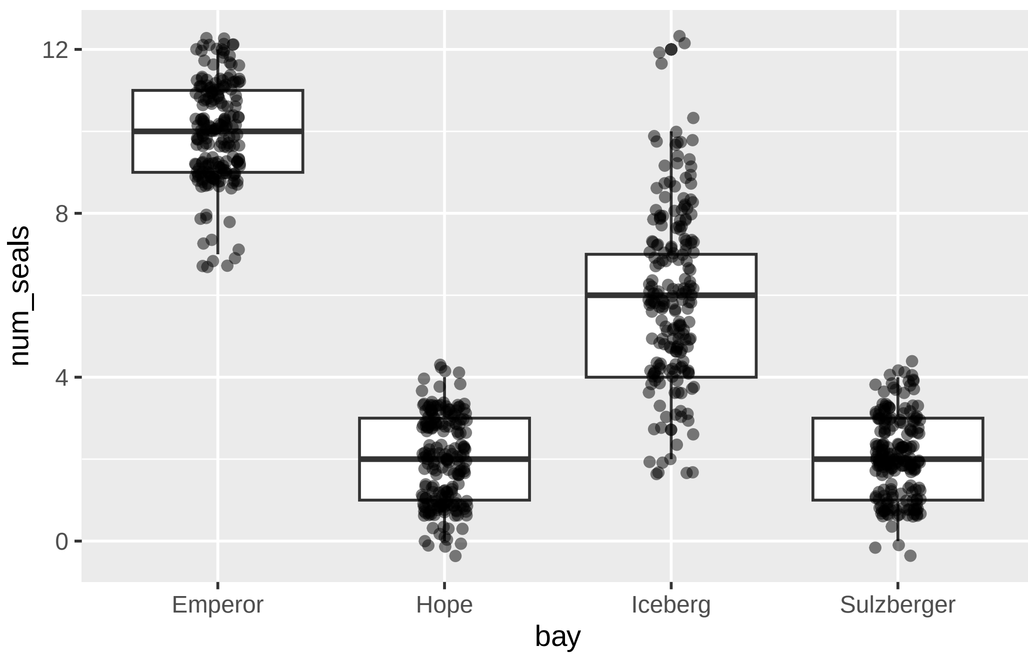
8.

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = fish_kg ~ bay, data = fish)

$bay
                          diff        lwr       upr     p adj
Hope-Emperor       -0.53796408 -2.2236167 1.1476886 0.8447466
Iceberg-Emperor     0.94195591 -0.7436968 2.6276086 0.4764075
Sulzberger-Emperor  0.04106038 -1.6445923 1.7267130 0.9999118
Iceberg-Hope        1.47991999 -0.2057327 3.1655727 0.1085321
Sulzberger-Hope     0.57902446 -1.1066282 2.2646771 0.8135811
Sulzberger-Iceberg -0.90089553 -2.5865482 0.7847571 0.5157262
```

11.

```
# A tibble: 4 x 3
  bay        mean_seal sd_seal
  <chr>          <dbl>   <dbl>
1 Emperor         9.96   1.19
2 Hope            1.96   1.02
3 Iceberg         6.02   2.13
4 Sulzberger      2.08   0.958
```

12.



13.

```
             Df Sum Sq Mean Sq F value Pr(>F)    
bay           3   8682    2894    1467 <2e-16 ***
Residuals   796   1570       2                   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

15.

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = num_seals ~ bay, data = seals)

$bay
                    diff        lwr        upr     p adj
Hope-Emperor       -8.00 -8.3615601 -7.6384399 0.0000000
Iceberg-Emperor    -3.94 -4.3015601 -3.5784399 0.0000000
Sulzberger-Emperor -7.88 -8.2415601 -7.5184399 0.0000000
Iceberg-Hope        4.06  3.6984399  4.4215601 0.0000000
Sulzberger-Hope     0.12 -0.2415601  0.4815601 0.8282111
Sulzberger-Iceberg -3.94 -4.3015601 -3.5784399 0.0000000
```